The Hessian Theorem in Resolution Geometry

Why Every Admissible Divergence Has the Fisher Hessian, and How Ignorance Converts into Error Bars

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Companion to *Resolution Geometry* and to *Čencov's Theorem in Resolution Geometry*

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Abstract

Small- and medium-scale measurement science works almost entirely with Hessians. These include:

- the curvature of a log-likelihood, which gives error bars;
- the curvature of a divergence, which gives distinguishability;
- the curvature of a free energy, which gives response and heat capacity;
- the curvature of a quantum entropy, which gives linear response and decoherence budgets.

We prove that one theorem stands behind all of them.

Theorem A (Hessian Čencov). Let a divergence vanish on the diagonal, be twice continuously differentiable there, be nondegenerate, and be monotone under all Markov maps. Then its Hessian is c times the Fisher metric, with one constant c in every dimension. The proof reduces the Hessian to a monotone metric and then applies Čencov's theorem as derived from RG descent. For f-divergences $c = f''(1)$, proved symbolically for generic f.

Theorem B (potentials and duality). Fisher is the Hessian of the negative entropy in mixture coordinates and of the log-partition function in natural coordinates. The two are Legendre-dual, so their Hessians are inverse. Kullback–Leibler divergence is the Bregman divergence of both.

Theorem C (conversion of ignorance). The information lost by a measurement chain converts *exactly* between four currencies:

- nats of relative-entropy loss;
- an integrated Hessian (Fisher) deficit;
- a bounded recovery error;
- a Cramér–Rao variance inflation.

In thermal families the Hessian is $k_B T^2 C_V$, so a heat-capacity measurement fixes the thermometry bound $\Delta T \geq T\sqrt{k_B/(nC_V)}$.

Theorem D (Cramér–Rao and Laplace). Variance bounds, nuisance Schur complements and posterior covariances are inverse Hessians.

Theorem E (quantum). Descent maps every monotone quantum divergence to a Petz monotone metric:

- BKM is the Hessian of the von Neumann negentropy;
- SLD is twice the Hessian of $-\log$ fidelity;
- WY is four times the Hessian of $-\log$ affinity.

Uniqueness is lost, so the task declares the Hessian.

Evidence. Every statement is either proved here, derived from the Čencov print, or explicitly imported. The verification battery is symbolic and deterministic, with no random sampling. Its witnesses are exact rational, symbolic, or 50-digit fixed instances.

Scope. Fisher–Rao, Hessian (dually flat), Bregman/Legendre, thermodynamic and quantum monotone geometries are thereby unified as *Hessians of descending potentials*. Each identity also becomes a laboratory protocol.

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1 Why a Hessian theorem, and why it matters in the laboratory

A laboratory never measures a metric tensor directly. It measures how fast a likelihood falls away from its maximum, how quickly two preparations become distinguishable, how a mean responds to a control, or how a heat capacity changes with temperature. Each of these is a second derivative. The engineering question is therefore which second derivative is the right one, and whether the answer depends on the choice of divergence, coordinates or protocol.

Resolution Geometry (RG) answers in two steps.

- The companion *Čencov's Theorem in RG* proves that an admissible metric, one that descends through lossless re-descriptions, is c·Fisher.
- Here we prove that the Hessian of every admissible divergence is such a metric (Theorem A).

Small-deviation geometry is therefore protocol-invariant up to one unit constant. Error ellipsoids computed from KL, χ^2, Hellinger or any other monotone smooth divergence coincide after rescaling. RG's "ignorance", meaning information made unavailable by a protocol, is then an exact Hessian deficit that converts into the quantities engineers budget: variance, recovery error and heat capacity.

Claim discipline. Classical facts are rederived for completeness and claimed as known: Cramér–Rao, the exponential-family identities, the Bregman form of KL, and the fluctuation identity. The proposed contributions are:

- the Hessian Čencov theorem obtained from RG descent;
- the conversion ledger;
- the quantum Hessian band;
- the protocols.

Their priority is unreviewed.

2 Setting

- **Laws and tangents.** Δ_n° denotes the positive laws on n outcomes, and $T_n = \{u : \sum_i u_i = 0\}$ the tangent space.
- **Fisher form.** $F_p(u) = \sum_i u_i^2/p_i$.
- **Channels.** A Markov map (channel) is a column-stochastic matrix K; outputs that never occur are deleted.
- **Divergences.** A *divergence* is a family $D = D_n : \Delta_n^\circ \times \Delta_n^\circ \to [0, \infty]$ with $D(p, p) = 0$. It is *monotone* (it descends through processing) if $D(Kp, Kq) \le D(p, q)$ for all channels K.
- **Diagonal Hessian.** Where it exists,

$$g_p^D(u) = \frac{d^2}{dt^2} D(p + tu, p)\Big|_{t=0}.$$

3 Theorem A: the Hessian Čencov theorem

Theorem 3.1 (Hessian Čencov)**.** *Let D be a divergence satisfying the following:*

(D1) $D(p, p) = 0$ *and* $D \ge 0$*;*

(D2) $t \mapsto D(p + tu, p)$ *is* C^2 *near* $t = 0$*, and* $p \mapsto g_p^D(u)$ *is continuous;*

(D3) D *is monotone under every Markov map;*

(D4) $g_p^D(u) > 0$ *for* $u \ne 0$*.*

Then $g^D = c\,F$ *with* $c = g_{U_2}^D((1, -1))/4 > 0$*, in every dimension and at every interior law. In particular every such divergence has the same small-deviation geometry up to one positive constant.*

Proof. Step 1: the expansion starts at second order. By (D1), $t = 0$ minimizes $t \mapsto D(p+tu, p)$. By (D2) its first derivative vanishes there, so $D(p+tu, p) = \frac{1}{2} g_p^D(u) t^2 + o(t^2)$.

Step 2: the Hessian is a monotone metric. Fix a channel K. By linearity $K(p+tu) = Kp + tKu$, and by (D3)

$$D(Kp + tKu,\, Kp) \le D(p + tu,\, p).$$

Divide by $t^2/2$ and let $t \to 0$ using Step 1 on both sides: $g^D_{Kp}(Ku) \le g^D_p(u)$. So g^D is a family of quadratic forms that is monotone under all Markov maps. It is continuous by (D2) and positive by (D4).

Step 3: monotone implies descent. If K has a Markov left inverse M, then monotonicity in both directions gives $g^D_{Kp}(Ku) = g^D_p(u)$. By the companion's Lemma, these K are exactly the congruent splittings, so g^D is an RG information metric.

Step 4: Čencov. By Čencov's theorem from RG descent (companion, Theorem 3.4), $g^D = cF$ with the stated c. Polarization extends the identity to the bilinear form. □

Corollary 3.2 (f-divergences)**.** *For $D_f(p,q) = \sum_i q_i f(p_i/q_i)$ with f convex, $f(1) = 0$ and f of class C^2 near 1,*

$$g^{D_f} = f''(1)\, F.$$

Proof. Directly,

$$\frac{d^2}{dt^2} \sum_i p_i f\left(1 + \frac{tu_i}{p_i}\right)\Big|_0 = f''(1) \sum_i \frac{u_i^2}{p_i},$$

and the first derivative is $f'(1)\sum_i u_i = 0$. This is verified symbolically for a generic f (E1). It is consistent with Theorem A, since every f-divergence is monotone. □

Examples. KL has $c = 1$; the Pearson χ^2 divergence has $c = 2$; the squared Hellinger distance $\sum(\sqrt{p} - \sqrt{q})^2$ has $c = \frac{1}{2}$; the Rényi divergence of order α has $c = \alpha$.

No-Go 3.3 (Each hypothesis is needed).

- *Without (D3).* The squared Euclidean divergence has Hessian $2I$, which is not a multiple of Fisher, and it is not monotone. Merging two outcomes of $p = (\frac{1}{6}, \frac{5}{12}, \frac{5}{12})$ and $q = (\frac{2}{3}, \frac{1}{6}, \frac{1}{6})$ raises it from $\frac{3}{8}$ to $\frac{1}{2}$ (E7, exact).
- *Without (D2).* Total variation is monotone but grows linearly, so it has no Hessian. Its local geometry is not quadratic.
- *Without (D4).* The zero divergence satisfies (D1)–(D3) but is degenerate.

Corollary 3.4 (Protocol-invariant error ellipsoids)**.** *For a smooth parametric model p_θ and any two divergences satisfying (D1)–(D4), the local ellipsoids $\{\delta\theta : D(p_{\theta+\delta\theta}, p_\theta) \le \epsilon\}$ coincide to second order after rescaling ϵ by the ratio of their constants c. The pulled-back tensor is $c\, J^\top F J$, where $J = \partial p/\partial\theta$.*

4 Theorem B: Hessian potentials and Legendre duality

Theorem 4.1 (Two potentials, one metric)**.** *(i)* Mixture potential. *With $\varphi(p) = \sum_i p_i \log p_i$, the Hessian satisfies $\operatorname{Hess}\varphi_p(u,u) = F_p(u)$ on T_n.*

(ii) Natural potential. *For an exponential family $p_\theta(x) = h(x)\exp(\theta \cdot T(x) - \psi(\theta))$,*

$$\operatorname{Hess}\psi(\theta) = \operatorname{Cov}_\theta(T) = F(\theta),$$

the Fisher matrix in natural coordinates.

(iii) Duality. *The mean map $\eta = \nabla\psi(\theta)$ is a diffeomorphism onto its image. The Legendre dual $\psi^*(\eta) = \sup_\theta[\theta \cdot \eta - \psi(\theta)]$ satisfies $\operatorname{Hess}\psi^*(\eta) = [\operatorname{Hess}\psi(\theta)]^{-1}$, and on the full categorical family ψ^* is φ restricted to mean coordinates.*

(iv) Bregman form of KL. *With* $B_\Phi(a,b) = \Phi(a) - \Phi(b) - \nabla\Phi(b)\cdot(a-b)$,

$$\mathrm{KL}(p\|q) = B_\varphi(p,q), \qquad \mathrm{KL}(p_\theta\|p_{\theta'}) = B_\psi(\theta',\theta).$$

Proof. (i) $\partial^2\varphi/\partial p_i\partial p_j = \delta_{ij}/p_i$.

(ii) Differentiating $\int h\, e^{\theta\cdot T-\psi} = 1$ once gives $\nabla\psi = \mathbb{E}_\theta T$, and twice gives $\operatorname{Hess}\psi = \operatorname{Cov}_\theta T$. The score is $T - \mathbb{E}_\theta T$, so its covariance is the Fisher matrix.

(iii) $\operatorname{Hess}\psi \succ 0$ makes $\nabla\psi$ injective with an invertible differential. The derivative of the Legendre transform inverts the gradient map, so the Hessians are inverse.

(iv) For (iv), expand $\varphi(p)-\varphi(q)-\sum_i(\log q_i+1)(p_i-q_i) = \sum_i p_i\log(p_i/q_i)$, using $\sum_i(p_i-q_i) = 0$. The natural form follows from $\log(p_\theta/p_{\theta'}) = (\theta-\theta')\cdot T-\psi(\theta)+\psi(\theta')$ and $\mathbb{E}_\theta T = \nabla\psi(\theta)$.
Items (i)–(iv) are verified symbolically on the categorical family with $n = 3$ (E2, E3). □

Proposition 4.2 (Three-point identity)**.** *For any convex* Φ,

$$B_\Phi(a,c) = B_\Phi(a,b) + B_\Phi(b,c) + (\nabla\Phi(b) - \nabla\Phi(c))\cdot(a-b).$$

When the last term vanishes, the Pythagorean relation of dually flat geometry holds. This happens when $a-b$ *is orthogonal to* $\nabla\Phi(b)-\nabla\Phi(c)$*, for instance at an m-projection onto an e-flat family. The full dual-connection structure is imported from Amari–Nagaoka [2].*

Proof. Expand all three Bregman terms; the potentials cancel pairwise. □

5 Theorem C: conversion of ignorance

Definition 5.1 (Ignorance of a protocol)**.** For a channel K, a law p and a reference q, the *ignorance* (information loss) is

$$L(p,q;K) = \mathrm{KL}(p\|q) - \mathrm{KL}(Kp\|Kq) \;\geq 0.$$

At a tangent u at p, the *Hessian deficit* is $\Delta_p(u) = F_p(u) - F_{Kp}(Ku) \geq 0$.

Theorem 5.2 (Exact exchange between currencies)**.** *Put* $v = p - q$ *and* $p_t = q + tv$.

(C1) Nats from Hessian deficit:

$$L(p,q;K) = \int_0^1 (1-t)\,\Delta_{p_t}(v)\,dt.$$

(C2) Hessian deficit as variance: $\Delta_p(u) = \mathbb{E}\big[(s - \mathbb{E}[s \mid Y])^2\big]$, *where* $s = u/p$ *is the score and* Y *the channel output.*

(C3) Recovery error from nats: *the reference-Bayes map* $R_{q,K}(x|y) = q_x K_{yx}/(Kq)_y$ *satisfies*

$$\mathrm{TV}(p, R_{q,K}Kp) \leq \sqrt{L/2}.$$

(C4) Variance inflation from deficit: *for a scalar parameter with input and output Fisher information* $F_{\mathrm{in}} \geq F_{\mathrm{out}} > 0$*, the Cramér–Rao bound inflates by*

$$\frac{1}{F_{\mathrm{out}}} - \frac{1}{F_{\mathrm{in}}} = \frac{\Delta}{F_{\mathrm{in}}F_{\mathrm{out}}} \;\geq\; \frac{\Delta}{F_{\mathrm{in}}^2}.$$

In the matrix case $F_{\mathrm{out}}^{-1} \succeq F_{\mathrm{in}}^{-1}$*. With nuisance parameters, the Schur complements obey the same ordering.*

(C5) Chains telescope: *for $p_j = K_j p_{j-1}$ and $q_j = K_j q_{j-1}$, the losses add, and the composed reference-Bayes recoveries equal the direct one.*

Proof.
- *(C1).* By Theorem 4.1(iv), KL $= B_\varphi$, and Taylor's formula with integral remainder gives $B_\varphi(p,q) = \int_0^1 (1-t)\,\mathrm{Hess}\,\varphi_{p_t}(v)\,dt$. Apply this to Kp, Kq with Kp_t, and subtract. An exact rational instance agrees to 40 digits (E4).
- *(C2).* Expand the square, using $\mathbb{E}[s \mid Y] = (Ku)/(Kp)$.
- *(C3).* Use the chain rule for relative entropy on the joint law, then Pinsker (companion, Theorem 6.4).
- *(C4).* Algebra, together with $F_{\mathrm{out}} \le F_{\mathrm{in}}$ from (C2). The matrix inequality follows from operator monotonicity of the inverse. For Schur complements see the companion, Proposition 10.5.
- *(C5).* Telescoping, together with the composition identity in the companion, Theorem 8.1. □

Theorem 5.3 (Thermodynamic currency)**.** *For a Gibbs family $p_\beta(x) \propto e^{-\beta E(x)}$:*

(i) the Fisher information about β is $F_\beta = \partial_\beta^2 \log Z = \mathrm{Var}(E) = k_B T^2 C_V$;

(ii) the Fisher information about T is $F_T = C_V/(k_B T^2)$;

(iii) hence any unbiased thermometer using n independent samples obeys

$$\Delta T \;\ge\; T\sqrt{\frac{k_B}{n\,C_V}}.$$

Proof. Theorem 4.1(ii) with $\theta = -\beta$ and $T(x) = E(x)$ gives (i). Differentiating $\langle E\rangle = -\partial_\beta \log Z$ with respect to T gives $C_V = \mathrm{Var}(E)/(k_B T^2)$. For (ii), use the chain rule with $d\beta/dT = -1/(k_B T^2)$. Then (iii) follows from Cramér–Rao (Theorem 6.1). Item (i) is verified symbolically for two- and three-level spectra (E6). □

Remark 5.4 (The conversion table).

Currency	Symbol	Exact relation
Information (nats)	L	$= \int_0^1 (1-t)\Delta_{p_t}(v)\,dt$ (C1)
Hessian deficit	Δ	= conditional score variance (C2)
Recovery error	TV	$\le \sqrt{L/2}$ (C3)
Variance	σ^2	$1/F_{\mathrm{out}} - 1/F_{\mathrm{in}} = \Delta/(F_{\mathrm{in}}F_{\mathrm{out}})$ (C4)
Heat capacity	C_V	$F_\beta = k_B T^2 C_V$; $\Delta T \ge T\sqrt{k_B/(nC_V)}$

Ignorance in RG's sense is therefore not a vague deficit. It is an exact Hessian deficit with fixed exchange rates into every unit a laboratory budgets.

6 Theorem D: Cramér–Rao and Laplace as inverse Hessians

Theorem 6.1 (Cramér–Rao, with nuisance)**.** *Let p_θ be regular and $\hat\theta$ unbiased for θ. Then*

$$\mathrm{Cov}(\hat\theta) \succeq F(\theta)^{-1}, \qquad F(\theta) = \mathbb{E}[-\,\mathrm{Hess}_\theta \log p_\theta] = \mathbb{E}[ss^\top].$$

For a target block a with nuisance block b, the target variance is bounded below by $(F_{aa} - F_{ab}F_{bb}^{-1}F_{ba})^{-1}$. In exponential families the bound is attained by the sample mean of T in mean coordinates.

Proof. Information identity. Differentiating $\int p_\theta = 1$ twice gives $\mathbb{E}[-\operatorname{Hess} \log p] = \mathbb{E}[ss^\top]$.

Scalar bound. Differentiate $\mathbb{E}\hat{\theta} = \theta$ to get $\operatorname{Cov}(\hat{\theta}, s) = 1$. Cauchy–Schwarz then gives $\operatorname{Var}(\hat{\theta}) \geq 1/F$. The matrix form applies this to every linear combination.

Nuisance. Invert the block matrix.

Attainment. The score is linear in T, so equality holds in Cauchy–Schwarz. For Bernoulli, the sample mean has variance $\theta(1-\theta)/n = 1/(nF)$ (E5, symbolic). □

Imported Theorem 6.2 (Laplace / Bernstein–von Mises [7])**.** *Under standard regularity, the posterior is asymptotically $\mathcal{N}(\hat{\theta}, \hat{J}^{-1})$, where $\hat{J}$ is the observed Hessian of the negative log-likelihood, and $\hat{J}/n \to F$.*

Both frequentist and Bayesian error bars are inverse Hessians of the same descending potential.

7 Theorem E: quantum Hessians, a band instead of a point

Theorem 7.1 (Quantum Hessian Čencov)**.** *Let D be a divergence on faithful density matrices that satisfies (D1)–(D4) with "Markov map" replaced by "CPTP map". Then its Hessian is a CPTP-monotone Riemannian metric, hence a member of the Petz family [3]. Unlike the classical case, different divergences give* different *members. Every member coincides with Fisher on commuting tangents.*

Proof. Steps 1–2 of Theorem 3.1 carry over verbatim, with CPTP maps in place of Markov maps, so the Hessian is a monotone metric. Petz's classification (imported) identifies the family. Commuting tangents reduce to the classical case. □

Proposition 7.2 (Named Hessians; executed, with imports)**.**

- BKM *is the Hessian of the von Neumann negentropy, and hence of the Umegaki relative entropy (companion, Theorem 13.4). For quantum Gibbs states it is the Kubo–Mori canonical correlation, i.e. linear response [6].*
- SLD *is* $2\times$ *the Hessian of* $-\log F$*, where* $F = (\operatorname{tr}|\sqrt{\rho}\sqrt{\sigma}|)^2$*. It is the quantum Cramér–Rao metric [5].*
- WY *is* $4\times$ *the Hessian of* $-\log \operatorname{tr}(\rho^{1/2}\sigma^{1/2})$.

On the fixed qubit $\rho = \begin{pmatrix} .7 & .1 \\ .1 & .3 \end{pmatrix}$ *with tangent* $A = \begin{pmatrix} .2 & .3 \\ .3 & -.2 \end{pmatrix}$*, in 50-digit arithmetic, the ratios of Hessian to metric are* 1, $\frac{1}{2}$ *and* $\frac{1}{4}$ *respectively (E8). The general correspondence between monotone metrics and quasi-entropy Hessians is imported [4].*

Remark 7.3 (Quantum thermometry). For $\rho_\beta \propto e^{-\beta H}$, the tangent $\partial_\beta \rho$ commutes with ρ. Every Petz metric then reduces to $\operatorname{Var}_\beta(H) = k_B T^2 C_V$, and the bound of Theorem 5.3 holds unchanged. The band reappears only for tangents that do not commute with the state. This replaces the toy variance assumption of the GR–QM multi-scale paper by a measurable heat capacity.

8 One theorem behind several geometries

Geometry	Potential whose Hessian it is	Placement here
Fisher–Rao (Riemannian)	any admissible divergence	Theorem 3.1, forced by descent
Hessian / dually flat	negentropy and log-partition	Theorem 4.1; Legendre-dual inverse Hessians

Geometry	Potential whose Hessian it is	Placement here
Bregman / convex	φ, ψ	KL $= B_\varphi = B_\psi$; three-point identity
Thermodynamic (Weinhold–Ruppeiner)	$\log Z$; entropy in mean coordinates	Theorem 5.3: $F_\beta = k_B T^2 C_V$
Estimation (Cramér–Rao, Laplace)	negative log-likelihood	Theorem 6.1: inverse Hessians
Quantum monotone (Petz, Kubo–Mori)	quantum divergences	Theorem 7.1: a band; the task declares the member

9 Laboratory protocols

Protocol 9.1 (Error budget from the Hessian)**.** 1. Fit the model.

2. Compute the observed Hessian $\hat{J}$ of the negative log-likelihood, and its expectation F if a design study is needed.

3. Form the Schur complement over the nuisance block.

4. Report $\hat{J}^{-1}_{\text{target}}$ as the covariance.

By Corollary 3.4, choosing a different admissible divergence for goodness-of-fit changes only a scale.

Protocol 9.2 (Ignorance budget of a measurement chain)**.** 1. For each stage K_j (detector, digitizer, binning, filter), compute the Fisher deficit Δ_j at the working point, or the loss L_j between two calibration laws.

2. Losses telescope (C5).

3. Convert the total into variance inflation (C4) and into a bound on the achievable recovery error (C3).

4. A stage with $\Delta_j = 0$ on the target tangent is lossless for that task, and may be simplified or removed.

Protocol 9.3 (Thermometry and decoherence budgets from the heat capacity)**.** 1. Measure $C_V(T)$.

2. The Fisher information per sample is $C_V/(k_B T^2)$.

3. The thermometry floor is $T\sqrt{k_B/(nC_V)}$.

4. For proper-time decoherence of a body at temperature T, the short-time exponent uses the variance $\mathrm{Var}(H_I) = k_B T^2 C_V$. This is a measured quantity, not a model assumption.

Protocol 9.4 (Quantum task declaration)**.** Declare the task before computing a metric:

- estimation uses SLD, the quantum Cramér–Rao bound;
- linear response or thermodynamics uses BKM;
- skew information and coherence resources use WY.

On commuting data all three coincide with Fisher, so the declaration matters exactly for coherent tangents.

10 Placement in the Theorem Ladder

- **L28–L29.** Fisher is forced not only as a metric (Čencov print) but as the Hessian of *every* admissible divergence (Theorem 3.1).
- **L35 (data processing).** The loss is an exact integrated Hessian deficit (C1), with exchange rates (C2–C5).
- **L38–L41 (quantum).** Descending divergences produce the Petz band (Theorem 7.1), and L39's task lock declares the Hessian.
- **RG's ignorance ledger.** Ignorance is a Hessian deficit with fixed conversions into nats, recovery error, variance and heat capacity.
- **Companions.**
  - The spectral-identifiability moduli are inverse-Hessian error scales.
  - The GR–QM multi-scale paper's decoherence exponent is $k_B T^2 C_V$.

11 Verification battery, limits, and non-claims

	Statement	Form of verification
E1	f-divergence Hessian $= f''(1)F$; first derivative 0	symbolic, generic f, $n = 3$
E2	$\operatorname{Hess}\varphi = F$; $\operatorname{Hess}\psi = \operatorname{Cov} T$; Legendre inverse	symbolic, categorical $n = 3$
E3	$\mathrm{KL} = B_\varphi$	symbolic
E4	loss = integrated Hessian deficit	exact rational instance, residual 0 at 40 digits
E5	Cramér–Rao attainment (Bernoulli)	symbolic
E6	$\partial_\beta^2 \log Z = \operatorname{Var} E = k_B T^2 C_V$	symbolic, 2- and 3-level
E7	Euclidean divergence not monotone ($\frac{3}{8} \to \frac{1}{2}$)	exact rational
E8	BKM, SLD, WY as Hessians (ratios 1, $\frac{1}{2}$, $\frac{1}{4}$)	fixed qubit, 50 digits

There is no random sampling: the battery is symbolic or exact, and reruns are byte-identical. The proofs are the evidence; the battery checks transcription and constants.

Limits.

- Finite alphabets; the dominated extension follows the Čencov companion, Section 12.
- Regularity (D2) and nondegeneracy (D4) are hypotheses.
- Imported results:
  - Čencov from descent (companion);
  - Petz's classification;
  - Lesniewski–Ruskai;
  - Bernstein–von Mises;
  - Amari–Nagaoka dual connections;
  - Kubo linear response;
  - Braunstein–Caves.
- The continuity question of the Čencov companion is inherited.

Non-claims.

- No new classical identity: Theorems B and D and the thermodynamic relations are classical and rederived here.
- No unique quantum Hessian.
- No finite-sample efficiency beyond exponential families.
- No experimental data.

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